



UTMxHackathon'26

Official Case Study

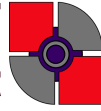
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ISSUED BY

Persatuan Mahasiswa Sains Komputer (PERSAKA),
Universiti Teknologi Malaysia (UTM)



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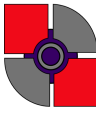
Welcome to UTMxHackathon'26!

This document presents 2 case studies across the domains of Agriculture and Retail.

Select **ONLY ONE** to solve, and remember to clearly state your chosen case study during your presentation.



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Case Study 1:

Precision Urban Agriculture for Vertical Farming (IoT System/ Web Dashboard/ Mobile Application)

1. Introduction

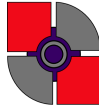
Rapid urbanization, shrinking arable land, and climate change are putting immense pressure on traditional food supply chains. In densely populated areas, food security and the carbon footprint of transporting produce are growing concerns. Vertical farming offers a space-efficient, sustainable alternative by growing crops in stacked layers within controlled indoor environments, making it possible to produce food directly within city limits

2. Problem Statement

Managing a vertical farm manually is highly resource-intensive and prone to human error. Inconsistent lighting, improper nutrient delivery, and fluctuating climate control can quickly lead to suboptimal crop yields, crop disease, wasted water, and excessively high energy consumption. Currently, many urban farmers and hobbyists lack access to affordable, data-driven automation tools needed to precisely monitor and optimize plant growth around the clock.

3. Challenge

Develop an intelligent, automated system that monitors and manages the environmental variables of an indoor vertical farming setup in real-time. The platform should collect sensor data, automate routine agricultural tasks, and provide actionable, data-driven insights to help users maximize crop yield while strictly minimizing water and electricity usage. It should be scalable enough for a small community garden or an indoor campus greenhouse.



4. What the solution should solve

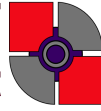
Participants can build a working hardware prototype (optional) paired with a software dashboard. Key features could include:

- Real-time monitoring using sensors for temperature, ambient humidity, soil moisture, and water pH levels.
- Automated control systems to adjust LED lighting spectrums, activate cooling fans, or trigger hydroponic water pumps based on specific plant profiles.
- A centralized web or mobile dashboard that visualizes growth metrics, resource consumption, and historical data trends.
- Predictive alerts that notify users of system anomalies, such as a water reservoir running dry or temperatures exceeding safe thresholds.
- An AI-driven recommendation engine that suggests optimal harvest times or adjustments to nutrient mixtures based on sensor inputs.

Area	Tools
Hardware / IoT	Arduino, ESP32, Raspberry Pi, Soil Moisture/pH Sensors
Backend	Node.js, Python (Flask/FastAPI), Firebase, Supabase
Frontend	React, Vue.js, Flutter, React Native
Data/AI	TensorFlow Lite, Scikit-learn, OpenAI API for crop insights
Communication	MQTT, WebSockets, REST APIs



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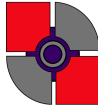
The solution should be:

- A functional hardware (optional) and software prototype demonstrating real-world data collection.
- Interactive and user-friendly, allowing individuals without deep agricultural expertise to manage a system.
- Scalable to adapt to different types of crops or larger vertical farming racks

Note: The features and tools listed above are provided as guidance only. Participants are strongly encouraged to explore alternative approaches that meaningfully address the problem statement.



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Case Study 2: The Youth Resilience Challenge

1. Introduction

Despite high internet penetration, a significant portion of Malaysian tertiary students and fresh graduates fall into debt traps early in their careers or find it hard to manage their budgets for future planning (e.g. saving and investing). While current banking tools provide transactional data, they largely lack the capacity for proactive, actionable intervention. This challenge invites participants to explore how technology can make "Financial Resilience" the default setting for a student's / early graduate's lifestyle, using GXBank's products as the foundational tool, can make financial resilience a default behaviour for students and early-career graduates, rather than an afterthought. Please feel free to recommend campaign ideas and/or new features to help this generation build positive financial habits

2. Problem Statement

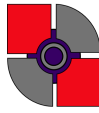
How can we leverage AI and behavioural economics to transform banking / savings habits from a passive storage of funds into an active, automated habit-builder that prevents debt accumulation among Malaysian youth?

3. Challenge

Design and build a solution that bridges the gap between financial awareness and consistent financial action for the next generation of Malaysian savers. The solution should go beyond data presentation and actively guide users toward healthier financial behaviour.



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Participants are expected to consider:

- How behavioural nudges (e.g. spending alerts, savings streaks, milestone rewards) can be embedded into daily banking interactions
- How AI can analyse spending patterns to generate personalised, context-aware financial guidance
- How automated savings mechanisms (e.g. round-up savings, salary-triggered transfers) can reduce reliance on user discipline
- How gamification or social accountability features could encourage long-term habit formation among the target demographic
- How the solution integrates with or proposes enhancements to existing GXBank product features

The solution may take the form of a mobile application, a feature concept with a working prototype, a campaign proposal supported by a functional demo, or any combination thereof. Proposals for new features or campaigns within the GXBank platform are also welcome.

Note: The examples above are provided as starting points.

Participants are encouraged to propose original, well-reasoned approaches that genuinely address the problem statement.

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